

Monolingual vs. Multilingual Fine-Tuning of AfroXLMR for Sentiment Analysis Across Five African Languages: A Code-Mixing Diagnostic Study

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Abstract

Sentiment analysis for African languages is constrained by limited annotated data, structural diversity, and pervasive English code-mixing. We compare four systems on five AfriSenti languages (Hausa, Yorùbá, Igbo, Nigerian Pidgin, Swahili): TF-IDF logistic regression, mBERT, and two fine-tuning regimes of AfroXLMR-base-Language-Adaptive Fine-Tuning (LAFT) and Multilingual Adaptive Fine-Tuning (MAFT). Our novel contribution is a token-level code-mixing index (CMI) filter partitioning each test set into monolingual and code-mixed subsets. LAFT achieves the highest weighted F1 on resource-rich Hausa (76.34); MAFT outperforms LAFT on low-resource Swahili (62.51 vs. 60.86), confirming a resource-driven crossover. Code-mixing effects are language-dependent, with Yorùbá improving on mixed tweets (+6.13 F1) while Hausa degrades modestly.

1 Introduction

Africa’s 2,000+ languages remain severely under-represented in NLP systems [1]. Sentiment analysis—the automatic classification of opinion as positive, negative, or neutral—is among the highest-demand downstream tasks, and its failure on African languages translates into absent or biased insights for over one billion speakers in health, political, and commercial domains.

The AfriSenti corpus [8] provides annotated sentiment data for 14 African languages, and AfroXLMR [2] extends XLM-R with African-language pretraining. Yet two questions remain open: (1) whether monolingual (LAFT) or joint multilingual (MAFT) fine-tuning of AfroXLMR is preferable, and whether this depends on resource level; and (2) whether code-mixing with English systematically degrades performance.

Our contributions are: (1) a controlled four-way comparison of TF-IDF, mBERT, LAFT, and MAFT across five typologically diverse languages; (2) a token-level CMI filter producing separate evaluations on monolingual and code-mixed test subsets; (3) empirical confirmation of a resource-driven LAFT/MAFT crossover; and (4) evidence that code-mixing effects are language-dependent. This work directly addresses African language underrepresentation and provides practitioners with evidence-based model-selection guidance.

2 Background

Muhammad et al. [6] introduced NaijaSenti, establishing sentiment annotation protocols and documenting pervasive code-mixing in Nigerian social media. They showed that standard multilingual models struggle particularly with Nigerian Pidgin due to its English creole structure. Muhammad et al. [8] extended this to AfriSenti (14 languages), and the AfriSenti-SemEval shared task [7] benchmarked 53 systems, with top results between 65–80 weighted F1, but without disaggregating by code-mixing rate.

Mabokela and Schlippe [4] and Mokhosi et al. [5] demonstrated that models trained on high-resource languages transfer poorly to morphologically rich Bantu and Sotho languages, motivating language-specific training strategies. Devlin et al. [3] introduced mBERT across 104 languages, but its African-language training data is negligible relative to European languages. Alabi et al. [2] introduced AfroXLMR, adapted via continued pretraining on 17 African languages, achieving state-of-the-art results on multiple African benchmarks.

Gap. No study has directly compared LAFT and MAFT of the same AfroXLMR model across multiple

Language	Train	Val	Test	Imb.
Hausa (hau, Afro-As.)	14,172	2,677	5,303	1.07×
Yorùbá (yor, NC)	8,522	2,090	4,515	1.89×
Igbo (ibo, NC)	10,192	1,841	3,682	1.73×
Nig. Pidgin (pcm, Cre.)	5,121	1,281	4,154	45.0×
Swahili (swa, NC)	1,810	453	748	5.61×

Table 1: Dataset statistics. Imbalance = max/min class count in training split. NC = Niger-Congo, Afro-As. = Afro-Asiatic, Cre. = English Creole.

language families on AfriSenti, nor applied a principled code-mixing diagnostic to evaluate performance degradation attributable to English code-mixing. We address both gaps.

3 Methodology

3.1 Dataset

We use AfriSenti [8], publicly available under Creative Commons licence. Five languages were selected to represent distinct typological families and resource levels (Table 1).

Labels are positive, neutral, or negative. Nigerian Pidgin’s extreme 45× imbalance (72 neutral vs. 3,241 negative training examples) and Swahili’s small training set (1,810) represent the primary data challenges.

3.2 Code-Mixing Diagnostic Filter

For each test tweet, we compute an English Token Ratio (ETR):

$$\text{ETR}(t) = \frac{|\text{English tokens}(t)|}{|\text{valid tokens}(t)|}$$

where valid tokens exclude @mentions, URLs, numerals, and tokens <3 characters. Per-token language detection uses `langdetect`. A tweet is code-mixed if $\text{ETR} > 0.20$. This threshold follows prior code-mixing literature and was not tuned on held-out data. Table 2 shows the resulting distribution.

3.3 Systems

TF-IDF + LR. Unigram/bigram TF-IDF (50k features, sublinear TF, diacritics preserved) with logistic regression (L2, $C=1.0$, balanced class weights). Non-neural lower bound.

mBERT. `bert-base-multilingual-cased` [3] fine-tuned per language. Standard cross-lingual reference.

Language	Test n	Mono.	Mixed (%)
Hausa	5,303	5,174	129 (2.4%)
Yorùbá	4,515	4,352	163 (3.6%)
Igbo	3,682	3,450	232 (6.3%)
Nigerian Pidgin	4,154	2,533	1,621 (39.0%)
Swahili	748	736	12 (1.6%)*

Table 2: Code-mixing diagnostic split. *Swahili mixed subset ($n=12$) is too small for robust evaluation.

Model	hau	yor	ibo	pcm	swa
TF-IDF+LR	69.84	72.64	77.35	65.06	57.83
mBERT	74.25	66.05	75.78	64.33	52.67
LAFT	76.34	70.27	77.24	68.02	60.86
MAFT	75.57	70.27	75.02	69.00	62.51

Table 3: Weighted F1 on full test sets. Bold = best per language.

LAFT. `Davlan/afro-xlmr-base` [2] fine-tuned independently on each language’s training set. Monolingual specialisation.

MAFT. `AfroXLMR-base` fine-tuned on the concatenation of all five training sets (39,817 examples), then evaluated per language. Tests cross-lingual parameter sharing.

3.4 Training Setup

All transformer models were trained with HuggingFace Trainer: learning rate 2×10^{-5} , batch size 32, linear schedule with 200 warmup steps, AdamW weight decay 0.01, up to 5 epochs with early stopping (patience = 2) on validation weighted F1. Maximum sequence length: 64 tokens (covers >99% of tweets). Single random seed (42). All experiments ran on NVIDIA T4 GPU.

3.5 Evaluation

Primary metric: weighted F1 (accounts for class imbalance). Secondary: macro F1 (exposes minority-class failure). Every system is evaluated on three subsets: full test, monolingual, code-mixed.

4 Experiments and Results

4.1 Main Results

Table 3 reports weighted F1 on the full test set.

RQ1 (LAFT vs. MAFT). LAFT achieves the highest weighted F1 on Hausa (76.34), the highest-

Lang.	Model	Mono.	Mixed	Δ
hau	LAFT	76.38	74.54	-1.84
hau	MAFT	75.65	72.22	-3.43
yor	LAFT	70.05	76.18	+6.13
yor	MAFT	68.80	78.87	+10.07
ibo	LAFT	76.93	81.48	+4.55
ibo	MAFT	74.80	78.10	+3.30
pcm	LAFT	65.96	71.25	+5.29
pcm	MAFT	66.68	72.64	+5.96
swa	LAFT	61.20	39.22	-21.98*
swa	MAFT	62.61	55.39	-7.22*

Table 4: Weighted F1 on monolingual and code-mixed subsets. Δ = Mixed - Mono. *Swahili mixed $n=12$.

resource language (14,172 training examples, $1.07 \times$ imbalance). MAFT outperforms LAFT on Swahili (62.51 vs. 60.86), the lowest-resource language (1,810 training examples). This confirms the resource-driven crossover: monolingual specialisation is advantageous when sufficient per-language data exists; cross-lingual parameter sharing compensates for data scarcity.

mBERT vs. TF-IDF. mBERT underperforms TF-IDF on Yorùbá (66.05 vs. 72.64). Yorùbá is tonal; lexical features may capture sentiment-bearing diacritics more reliably than subword tokenisation that fragments tonal characters. On Swahili, mBERT achieves only 52.67, well below the TF-IDF baseline (57.83), highlighting the limitations of generic multilingual pretraining for low-resource Bantu languages.

RQ2 (class imbalance). The weighted/macro F1 gap on Nigerian Pidgin is severe across all models. MAFT achieves 69.00 weighted F1 but only 50.95 macro F1—a 18-point gap driven by the 72-example neutral class being effectively ignored. Balanced class weights in TF-IDF and standard cross-entropy in transformers both fail to overcome a $45 \times$ training imbalance.

4.2 Code-Mixing Diagnostic

Table 4 reports the performance gap between monolingual and code-mixed subsets.

RQ3 (code-mixing effects). Code-mixing does not universally degrade performance. Yorùbá LAFT improves from 70.05 (mono) to 76.18 (mixed), a +6.13 gain. Igbo and Nigerian Pidgin also show improvements on code-mixed subsets. A plausible explanation:

English function words in code-mixed Yorùbá and Igbo tweets carry strong sentiment polarity that the model recognises, while purely African-language sentiment expressions involve tonal and orthographic ambiguity that is harder to classify from romanised text.

Hausa shows modest degradation (-1.84 for LAFT), consistent with its near-monolingual test distribution (2.4% code-mixed). Swahili’s severe LAFT degradation (-21.98) should be treated with caution given only 12 mixed-subset tweets. MAFT’s smaller Swahili degradation (-7.22) may indicate robustness from multilingual training, but cannot be confirmed from 12 examples.

4.3 Error Analysis

LIME explanations applied to TF-IDF misclassifications reveal four dominant error categories. Code-switching errors account for 40% of sampled Nigerian Pidgin misclassifications (12/30), where the model encounters English phrases not associated with surrounding Pidgin sentiment context. Orthographic variation (repeated characters, non-standard spelling) explains errors in Hausa (2/30) and Igbo (2/30). Short question-form tweets (<8 tokens with ‘?’) contribute sarcasm-related errors across all languages. The majority of remaining errors (labelled ‘other’) likely reflect label noise inherent in crowdsourced annotation of subjective sentiment.

4.4 Responsible NLP Reflection

Performance equity across languages is poor. Swahili’s best weighted F1 (62.51) is 13.83 points below Hausa (76.34)—a gap driven entirely by data availability, not linguistic complexity. The macro F1 gap on Nigerian Pidgin reveals that the neutral class is invisible to all models, biasing any deployed system toward binary polarisation.

5 Reflections and Discussion

What worked. The LAFT/MAFT crossover is theoretically coherent and empirically confirmed. TF-IDF’s competitive performance on Yorùbá (72.64, above mBERT at 66.05) suggests lexical baselines remain relevant for tonal languages, with practical implications for GPU-constrained deployment.

What did not work. Nigerian Pidgin’s neutral class was not rescued by any model. The macro F1

ceiling near 50 indicates that 72 neutral training examples are insufficient regardless of architecture. Data augmentation or targeted annotation is the most direct path to improvement.

Limitations. Single random seed (42) without variance estimation limits reproducibility confidence. The CMI filter is a heuristic with known noise on short tokens. Swahili’s 12-tweet mixed subset yields unreliable estimates. LIME was applied to TF-IDF only; transformer interpretability was not evaluated. Hyperparameters were fixed without grid search.

Responsible AI. All datasets are publicly available under Creative Commons licences. Code and results are fully reproducible from the provided notebooks and requirements files. AfriSenti was annotated by native speakers using majority-vote protocols, though crowdsourced annotation inherits subjectivity biases. Future work should extend to the full 14-language AfriSenti corpus and investigate data augmentation for severely imbalanced classes.

Extensions. Three directions follow naturally. First, language-balanced MAFT sampling (oversampling Swahili) may improve low-resource results while preserving high-resource performance. Second, extending the CMI diagnostic to the full AfriSenti corpus would provide sufficient sample sizes for robust Swahili analysis. Third, applying LAFT/MAFT comparison at AfroXLMR-large scale would test whether the crossover persists at greater model capacity.

6 Conclusion

We compared four sentiment systems across five AfriSenti languages using a novel code-mixing diagnostic that partitions test sets into monolingual and code-mixed subsets. Three findings emerge:

(1) LAFT outperforms MAFT on high-resource Hausa (76.34 vs. 75.57 wF1); MAFT outperforms LAFT on low-resource Swahili (62.51 vs. 60.86), confirming a resource-driven crossover (RQ1).

(2) Nigerian Pidgin’s $45\times$ class imbalance produces a severe wF1/mF1 gap that no model resolves, demonstrating that data quality constraints dominate architectural choices for severely imbalanced classes (RQ2).

(3) Code-mixing effects are language-dependent. Yorùbá and Igbo improve on code-mixed subsets, while Hausa degrades modestly. The Swahili result is inconclusive (RQ3).

For practitioners: LAFT is preferred for languages with $>10,000$ training examples; MAFT offers measurable advantage below this threshold. The CMI diagnostic is a reusable evaluation tool for any African language benchmark where English influence is present.

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